HIMANSHU PAITHANE

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WORK EXPERIENCE

Data Engineer – ML Analytics | PANASONIC NORTH AMERICA — ***May 2025 - Present***

- Developed and Deployed a **Databricks** (PySpark) pipeline to transform and standardize **300,000+** customer names using **NLP** and fuzzy matching, improving data quality, classification quality and reliability of downstream predictive analytics and production ready datasets.
- Built a scalable **PySpark** data validation framework with embedded **SQL** checks to automate **ETL** reconciliation and migration QA, improving data quality monitoring, data integrity and supporting **CI/CD** validation workflows for reliable pipeline testing and deployment.
- Automated **GCP** ETL pipelines into **BigQuery** for product engagement and lifecycle analytics, then integrated curated datasets into **Power BI** Dashboards to support KPI visibility, stakeholder reporting and product decision-making.

Data Scientist | JAR - Fintech ($300M Valuation) — ***Mar 2024 - Aug 2024***

- Developed, validated and deployed an **ML**-based affluence scoring model using gradient boosting frameworks - **LightGBM** and **XGBoost** to classify new users into spending-affinity tiers, improving targeting decisions while replacing a third-party solution - saving **~$15k/month**.
- Built data pipelines for streaming and batch-oriented data workflow using **Kafka**, **Airflow**, **AWS S3** and **MongoDB** to support large-scale analytics workflows and product insights on user event data from **~20K** daily active users.
- Applied **NLP** techniques to transform unstructured SMS data into structured datasets for behavioral analysis, trend identification, user segmentation and engagement reporting for downstream analysis, pattern recognition and predictive modeling insights through **Tableau** dashboard reporting.

Data Science Intern | SYMPHONY AI — ***May 2023 - Aug 2023***

- Built and optimized machine learning models for **time-series forecasting** of motor health to identify exceptions and risk factors using sensor-derived time-series data, improving accuracy by **23%** through feature engineering, hyperparameter tuning, performance analysis and model evaluation.
- Improved accuracy of production ML models including Regression based models and **neural networks** through feature engineering, statistical analysis, and hyperparameter tuning, boosting **F1-score to 90%** and enabling more reliable predictions at scale.
- Worked across end-to-end ML lifecycle including data preprocessing, feature engineering, model training, validation, and production-oriented model improvement across diverse ML models to deliver reliable, production-oriented predictions and performance analysis.

Data Science Intern | SHARP — ***Jun 2021 - Sep 2021***

- Built end-to-end machine learning workflows using **Python** libraries including PyTorch, scikit-Learn, TensorFlow, Keras to deliver high-accuracy, production-ready predictive models.
- Designed, executed, and analyzed large scale **A/B tests** to evaluate product features and business strategies, applying statistical methods to generate actionable insights that directly guided product decisions and user engagement improvements.
- Optimized big data pipelines using **Apache Spark** & **Hadoop** to process large datasets, boosting efficiency by 4x and enabling scalable analytics and EDA.

PROJECTS (Check out more here)

VectorSearch RAG (MongoDB) - LangChain Chunking + Sentence-Transformers + PyMongo | Link

- Built an end-to-end RAG workflow for retrieval in Python using LangChain to chunk financial document PDFs, generate sentence-transformer embeddings, and store vectors and metadata in MongoDB Atlas to retrieve grounded context for LLM-powered question answering.
- Implemented retrieval and grounding workflows with MongoDB Vector Search queries to retrieve top semantically similar chunks to improve response relevance, s upport functional/tool-style retrieval steps, and reduce hallucination risk.
- Iterated on prompt design, retrieval configuration, and evaluation logic to improve output quality, latency, and reliability in practical AI-assisted workflows

Multimodal Document Parsing Pipeline - OpenAI Responses API, Vision, PyMuPDF, Pydantic | Link

- Built an LLM-based document processing and extraction and classification pipeline in Python using the OpenAI Responses API to convert PDF tables into structured records using prompt design, schema validation, and grounded extraction logic to export clean CSVs for analytics, reporting and validation.
- Optimized on prompt design, JSON schema logic and Pydantic validation to improve metadata extraction, classification accuracy and output consistency.
- Added multi-run evaluation checks, output validation and guardrail constraints to improve consistency, safely, and reliability of AI-generated outputs.

FindMySquad - Full-Stack Web Development | Link

- Developed and deployed a full-stack web platform (Node.js, Express.js, MongoDB, Handlebars, CSS, JavaScript) for hosting/joining games, finding gym buddies, and organizing esports tournaments.
- Integrated real-time chat (Websockets), email reminders (Nodemailer + Cron), and interactive maps (Leaflet.js) to enhance collaboration and event coordination.
- Designed and implemented user engagement systems (profiles, ratings, karma leaderboard, achievements) to boost retention and fair play.

SKILLS

- **Programming & Querying:** Python, SQL, R
- **Data Engineering:** ETL, ELT, Data Pipelines, Data Integration, Data Transformation, Data Modeling, Automated Workflows, Production Data Pipelines
- **Big Data & Orchestration:** Databricks, Apache Spark, PySpark, Kafka, Airflow, Hadoop
- **Databases & Cloud Platforms:** BigQuery, MongoDB, PostgreSQL, AWS S3, GCP
- **Data Quality & Reliability:** Data Validation, Data Quality Monitoring, Data Integrity, Reproducible Workflows, CI/CD Validation Workflows
- **Reporting & Visualization:** Power BI, Tableau, Dashboard Development, KPI Reporting, Data Visualization
- **Applied AI/ML:** LLMs, RAG, Embeddings, Vector Search, Retrieval, Grounding, Prompt Design, Evaluation, Guardrails, Data Science, Machine Learning

EDUCATION

STEVENS INSTITUTE OF TECHNOLOGY

Master of Science, Computer Science

- GPA**: 3.9/4.0**